Section 3.3 Diagonalization and Eigenvalues

Exercise 3.3.3

Show that A has $\lambda = 0$ as an eigenvalue if and only if A is not invertible.

Exercise 3.3.6

Find the characteristic polynomial of the $n \times n$ identity matrix I. Show that I has exactly one eigenvalue and find the eigenvectors.

Exercise 3.3.7

Given $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ show that:

a. $c_A(x) = x^2 - \operatorname{tr} Ax + \det A$, where $\operatorname{tr} A = a + d$ is called the trace of A.

Section 5.1 Subspaces and Spanning

Exercise 5.1.1

In each case determine whether U is a subspace of $\mathbb{R}^3$. Support your answer.

a. $U = \{(1, s, t) \mid s \text{ and } t \text{ in } \mathbb{R}\}$.

c. $U = \{(r, s, t) \mid r, s, \text{ and } t \text{ in } \mathbb{R}, -r + 3s + 2t = 0\}$

f. $U = \{(2r, -s^2, t) \mid r, s, \text{ and } t \text{ in } \mathbb{R}\}$.

Exercise 5.1.3

In each case determine if the given vectors span $\mathbb{R}^4$. Support your answer.

a. $\{(1,1,1,1),(0,1,1,1),(0,0,1,1),(0,0,0,1)\}$.

Exercise 5.1.4

Is it possible that $\{(1, 2, 0), (2, 0, 3)\}$ can span the subspace $U = \{(r, s, 0) \mid r \text{ and } s \text{ in } \mathbb{R}\}$? Defend your answer.

Exercise 5.1.12

Suppose that $U = \text{span}\{x_1, x_2, \ldots, x_k\}$ where each x_i is in $\mathbb{R}^n$. If A is an $m \times n$ matrix and $Ax_i = 0$ for each i, show that $Ay = 0$ for every vector y in U.

Exercise 5.1.22

Let U and W be subspaces of $\mathbb{R}^n$. Define their intersection $U \cap W$ and their sum $U + W$ as follows: $U \cap W = \{x \in \mathbb{R}^n \mid x \text{ belongs to both } U \text{ and } W\}$. $U + W = \{x \in \mathbb{R}^n \mid x \text{ is a sum of a vector in } U \text{ and a vector in } W\}$.

a. Show that $U \cap W$ is a subspace of $\mathbb{R}^n$.

b. Show that $U + W$ is a subspace of $\mathbb{R}^n$.

Section 5.2 Independence and Dimension

Exercise 5.2.1

Which of the following subsets are independent? Support your answer.

b. $\{(1,1,1),(1,-1,1),(0,0,1)\}$ in $\mathbb{R}^3$.

d. $\{(1,1,0,0),(1,0,1,0),(0,0,1,1),(0,1,0,1)\}$ in $\mathbb{R}^4$.

Exercise 5.2.2

Let $\{x, y, z, w\}$ be an independent set in $\mathbb{R}^n$. Which of the following sets is independent? Support your answer.

b. $\{x + y, y + z, z + x\}$

Exercise 5.2.4

Find a basis and calculate the dimension of the following subspaces of $\mathbb{R}^4$.

b. $U = \left\{ \begin{bmatrix} a-b \\ b+c \\ b+c \\ b+c \end{bmatrix} \;\middle|\; a, b, \text{ and } c \text{ in } \mathbb{R} \right\}$

e. $U = \left\{ \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} \;\middle|\; a+b-c+d = 0 \text{ in } \mathbb{R} \right\}$